4. Files and troubleshooting

4.1 Where everything is kept

What	Where
Installed programs	`/usr/local/bin/treesync`, `/usr/local/bin/treesync-cli` (`root:root`, 755; by `install.sh`)
Installed manual	`/usr/local/share/doc/treesync/pdf/`, and `INSTALLED`: which build, commit, when
Desktop entry	`/usr/local/share/applications/treesync.desktop`
polkit action	`/usr/share/polkit-1/actions/org.treesync.run-as-root.policy`
Built programs	`v001/out/gcc/` and `v001/out/clang/`: `treesync`, `treesync-cli`, `treesync-test`
Build and test output	`v001/out/` (`verify/` logs, `bootstrap-*.txt`); ignored by git
GUI settings	`~/.config/treesync/settings`: one option per line, as on the command line
Build page choices	`~/.config/treesync/build`
Database	`~/.local/share/treesync/treesync.db` (SQLite): runs, actions, hash cache, checks
Run logs	`~/.local/share/treesync/logs/YYYY-MM-DD_HHMMSS.log`
Results of every run	`~/.local/share/treesync/logs/Results.txt`, one line per run
AI exports	`~/.local/share/treesync/logs/ai/run-N-logs.AI.md`, `run-N-sql.AI.md`
Root runs	the same under `/root/.local/share/treesync/`
Safe-delete folder	`<target>.old` next to the target (or `--safe-delete-dir`)
Unfinished copies	`NAME.treesync-part` next to where the file goes; removed when done
This manual	`Manual/*.md` (the sources) and `Manual/pdf/` (built by `Manual/build-pdf.sh`)

`--db=PATH` puts the database (and its `logs/` folder) somewhere else.

4.2 Troubleshooting

"TreeSync is not running as root ... only inside your home directory"

That is the source-tree build running as you. Use the installed TreeSync, which syncs as root for administrators (`./install.sh`, then **TreeSync** in the app list or `treesync-cli ...`), or tick **Sync > Run as root**. Inside your home nothing is needed.

Root runs use an older TreeSync, or the log says the installed one "is not this build"

The installed copy does not follow `out/`: run `./install.sh` (or **File > Install for root**) after building. `cat /usr/local/share/doc/treesync/INSTALLED` shows which build is installed.

"ask an administrator (a member of wheel) to run TreeSync as root"

You are not an administrator, so TreeSync syncs as you, inside your home. To sync other folders, an administrator runs it, or adds you to wheel (`sudo usermod -aG wheel NAME`; log in again), which also lets you use sudo.

The password is asked at every run

The polkit action is missing: it is installed only when `install.sh` runs as root (the default, through sudo), not for a `--prefix` of your own.

`sudo treesync-cli`: "command not found"

sudo does not search `/usr/local/bin` on RHEL. Leave the sudo out: the installed `treesync-cli` starts itself through sudo.

TreeSync is not in the app list

Run `./install.sh`; log out and in again if the desktop does not pick it up at once.

A path is refused

Without root, a path that leaves your home (also through `..` or a symlink) is refused. Use root, or a path inside your home.

bootstrap stops: "Still missing: ..."

A required package did not install. Run the install command it printed by hand and read its error (often a repository that is not enabled, for example CodeReady Builder on RHEL for `gtk3-devel`), then run `./bootstrap.sh` again.

build.sh: "U++ (umk + uppsrc) not found"

Run `./bootstrap.sh`, or point to an existing U++ with `export UPP_DIR=/path/to/u++`.

Building as root fails: "U++ not found", or umk "error while loading shared libraries"

Older scripts looked for U++ in root's home and then picked a U++ built for another distribution. Current ones use the U++ of the folder's owner (`~owner/u++`) and skip one that cannot run here. Root-owned leftovers in `out/` from such a build: `sudo chown -R $USER v001/out`.

A gcc release build crashes at start

The U++ heap fix is missing (`uppfix`); `./bootstrap.sh` applies it.

A test fails

`./verify.sh` stops and prints the failing checks; the full output is in `v001/out/verify/<compiler>-<type>.log`. `git push --no-verify` pushes anyway, but fix it first if you can.

A run is slow

Make sure the GUI runs a **release** build (`./verify.sh` leaves them in `out/`). Many small files: more threads (`--threads=8`) can help. Over SFTP the network is usually the limit; `--compress` helps text, not media.

"no Wayland session found" on the Build page
The GUI was started without `WAYLAND_DISPLAY`, usually through `sudo`. Start it as yourself and use **Sync > Run as root** for root runs.

The host key is refused (remote trees)
Connect once with `ssh user@host` to check and accept the key; TreeSync never accepts an unknown or changed key by itself.

A run was stopped and the target has `.treesync-part` files
Run the same sync again: it resumes, re-checks what was already copied, and removes the part files when done. `--fresh` starts over instead.

The `.old` folder keeps growing
Safe delete never deletes. **Sync > Clear .old folders...** (or `treesync-cli --clear-old --no-dry-run`) empties it when you are sure.

Something else
View > Export newest run for AI and give the files in `logs/ai/` to an AI, or read them yourself: they list every problem of the run first.